# Computation and Communication Ceilings in Roofline Model

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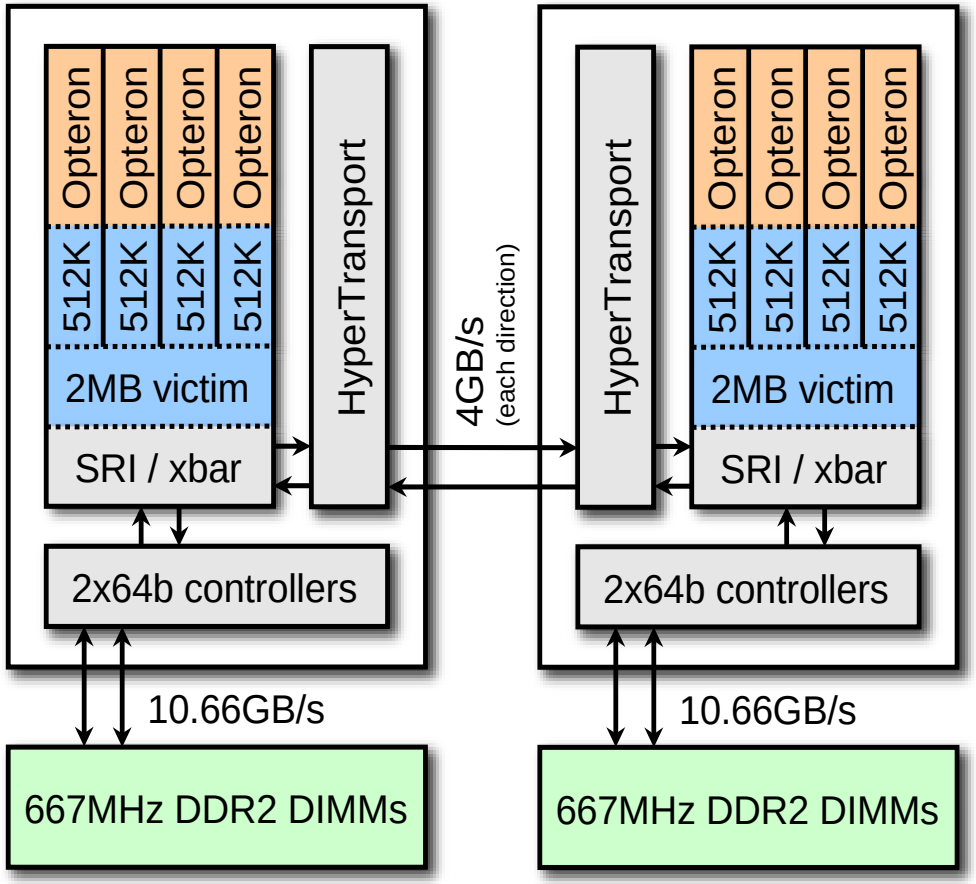
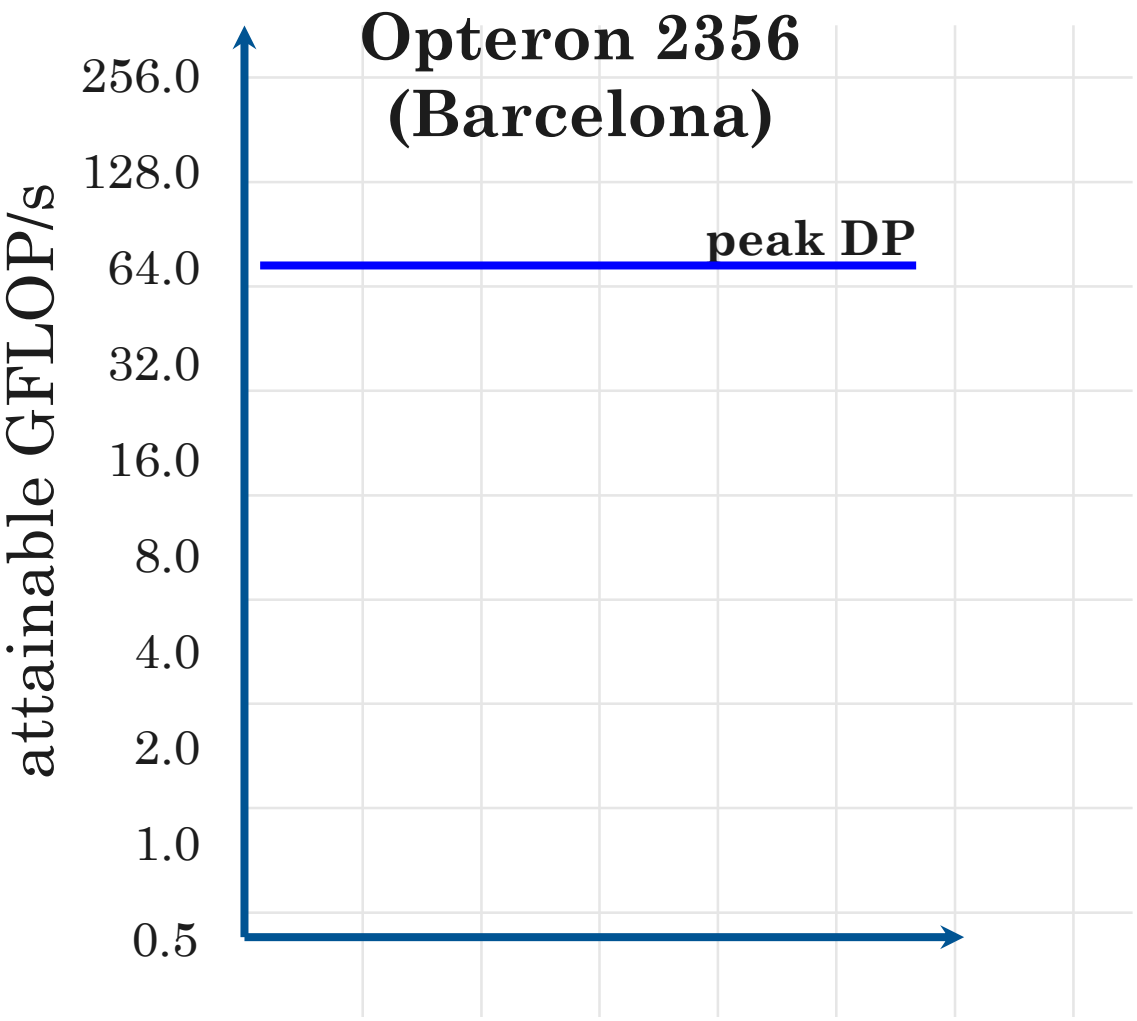
# Example of Opteron Processor

- Consider the Opteron 2356:
  - Dual Socket (NUMA)
  - Limited HW stream prefetchers
  - Quad-core (8 total)
  - 2.3GHz
  - 2-way SIMD (DP)
  - Separate FPMUL and FPADD datapaths
  - 4-cycle FP latency



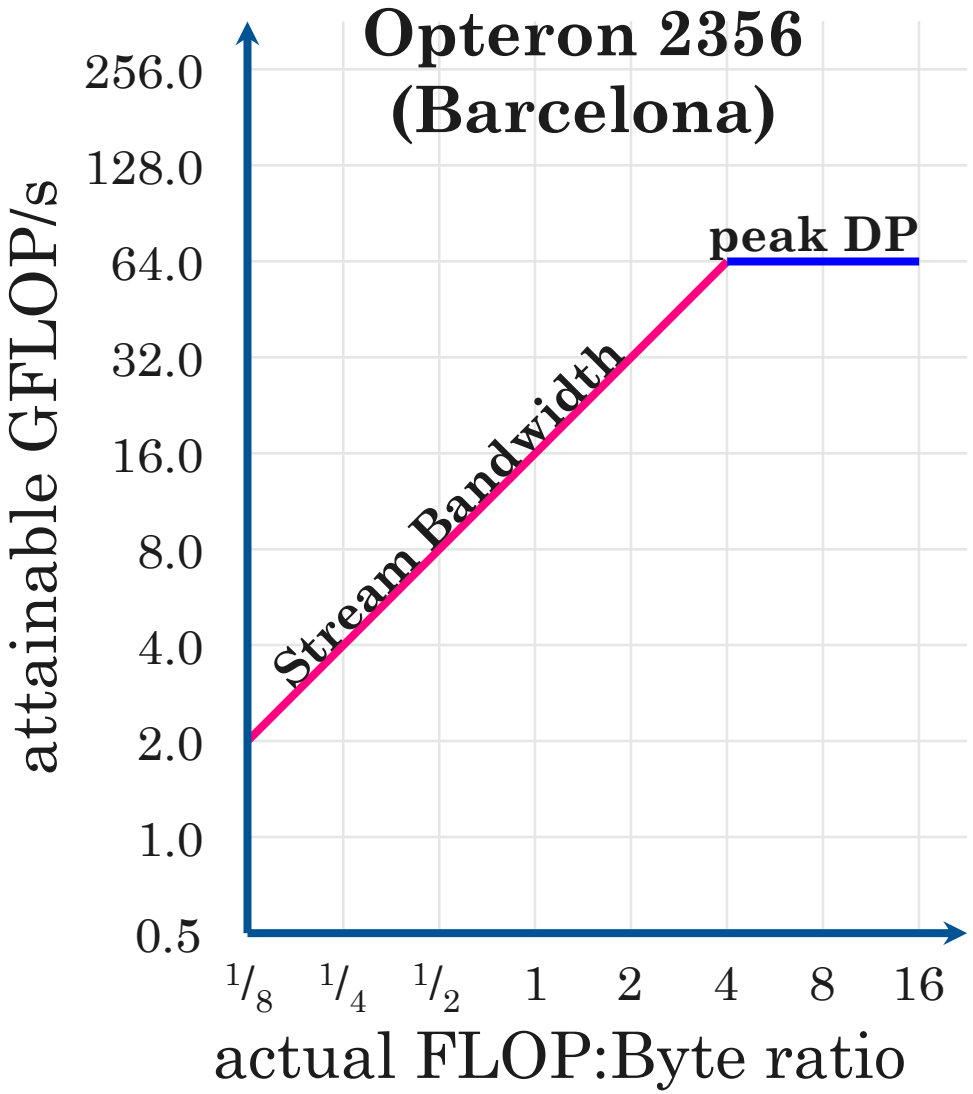

- Let us study its roofline model

# Roofline Model
## Basic Concept




- Naively, one might assume peak performance is always attainable.

# Roofline Model
## Basic Concept

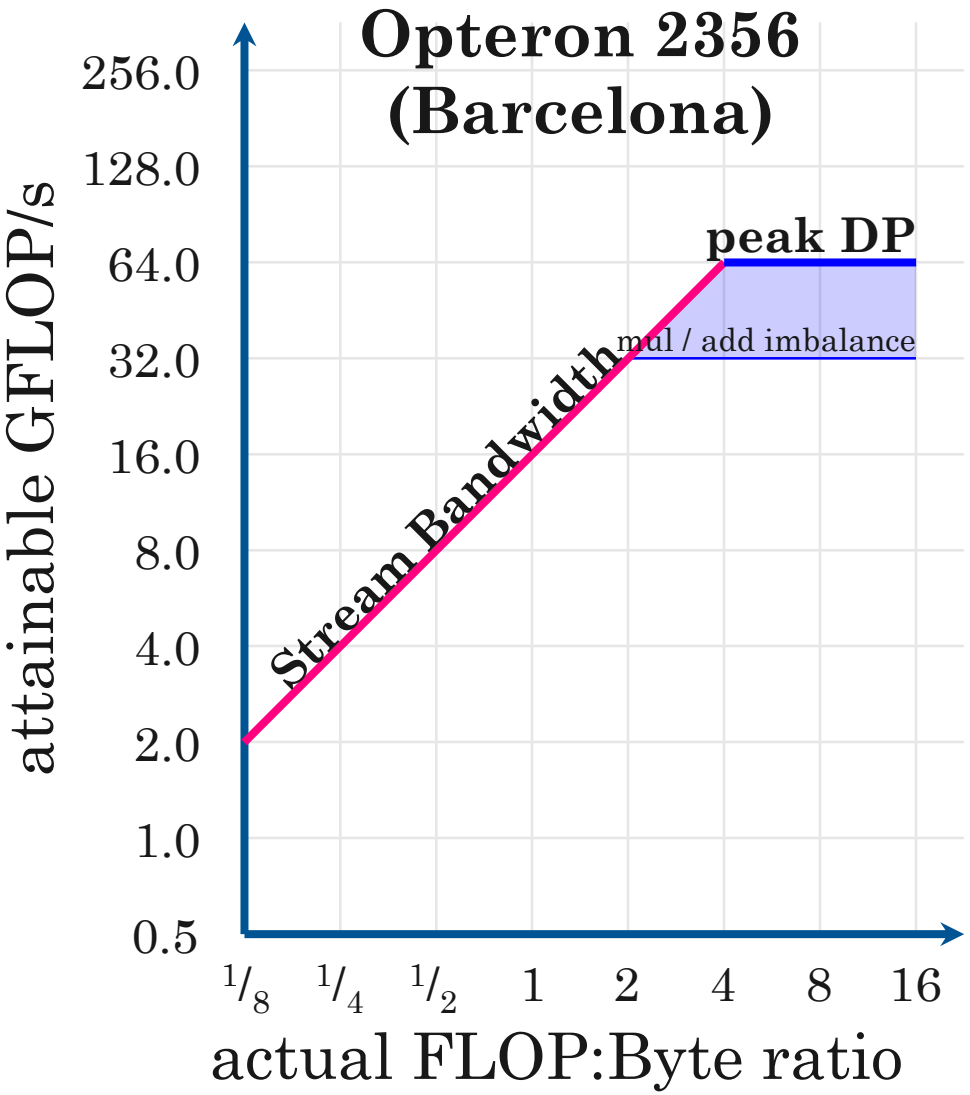



- However, with a lack of locality, **DRAM bandwidth can be a bottleneck**
- Given AI, we can easily bound performance
- But architectures are much more complicated
- We will bound performance as we eliminate specific forms of in-core parallelism

# Computation Ceilings

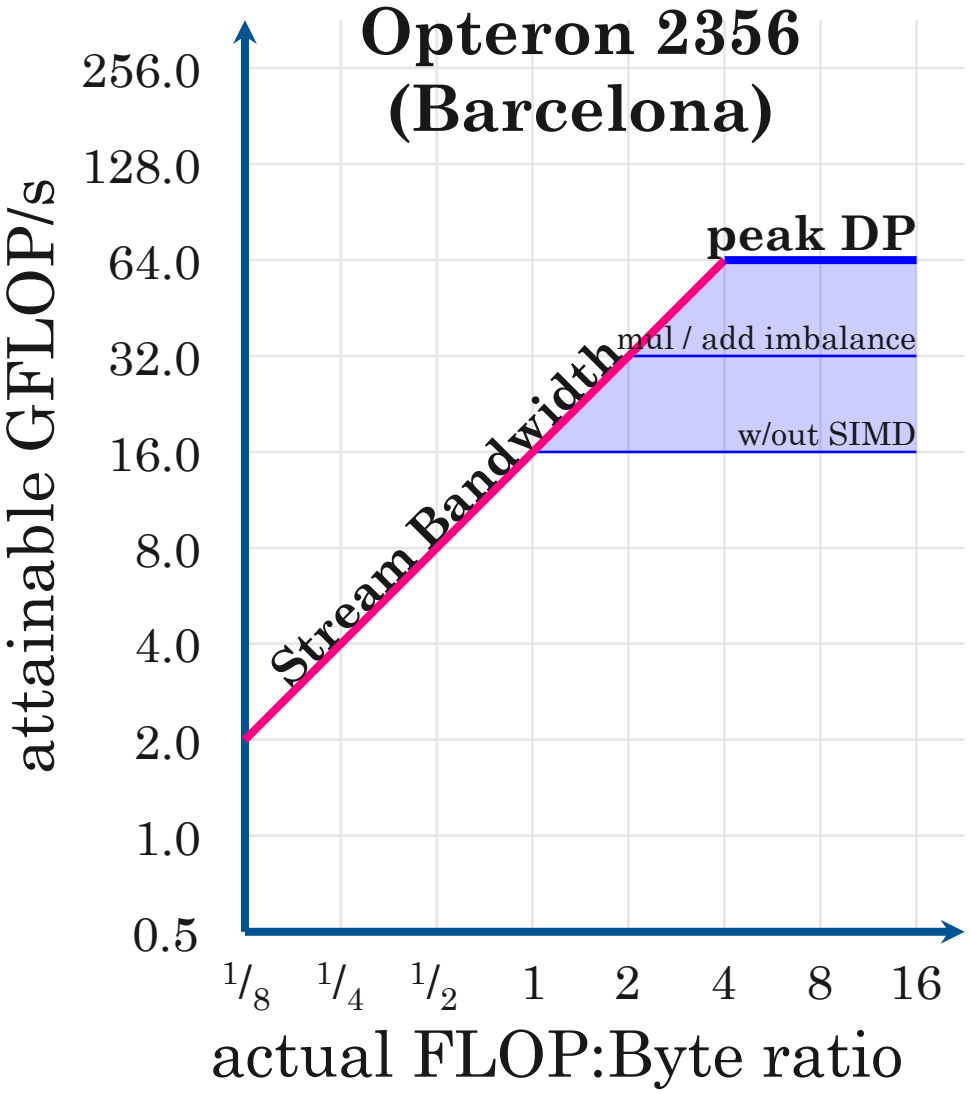
# Roofline Model

## Computational Ceilings



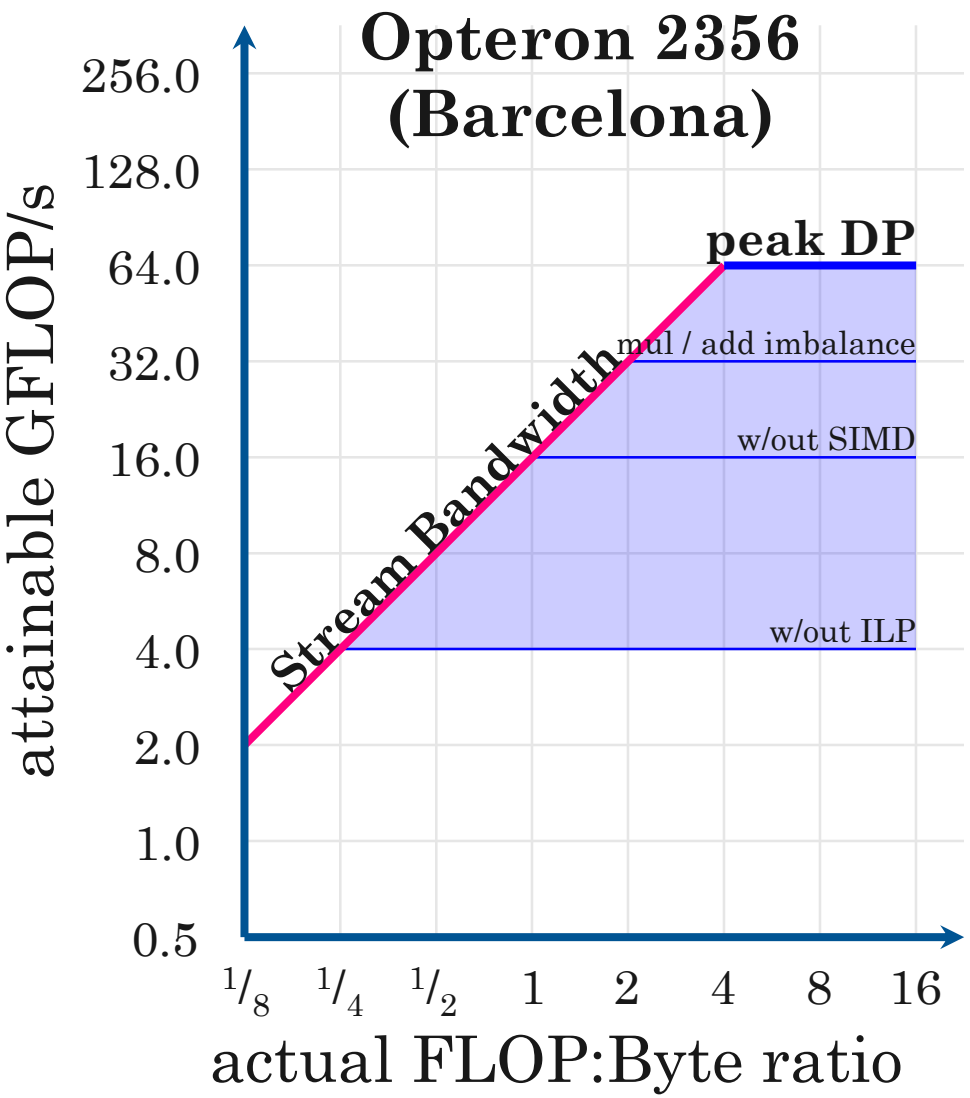

- Opterons have dedicated multipliers and adders.
- If the code is dominated by adds, then attainable performance is half of peak.
- We call these **Ceilings**
- They act like constraints on performance

# Roofline Model
## Computational Ceilings




- Opterons have 128-bit datapaths.
- If we cannot leverage SIMD, attainable performance will be halved

# Roofline Model
## Computational Ceilings

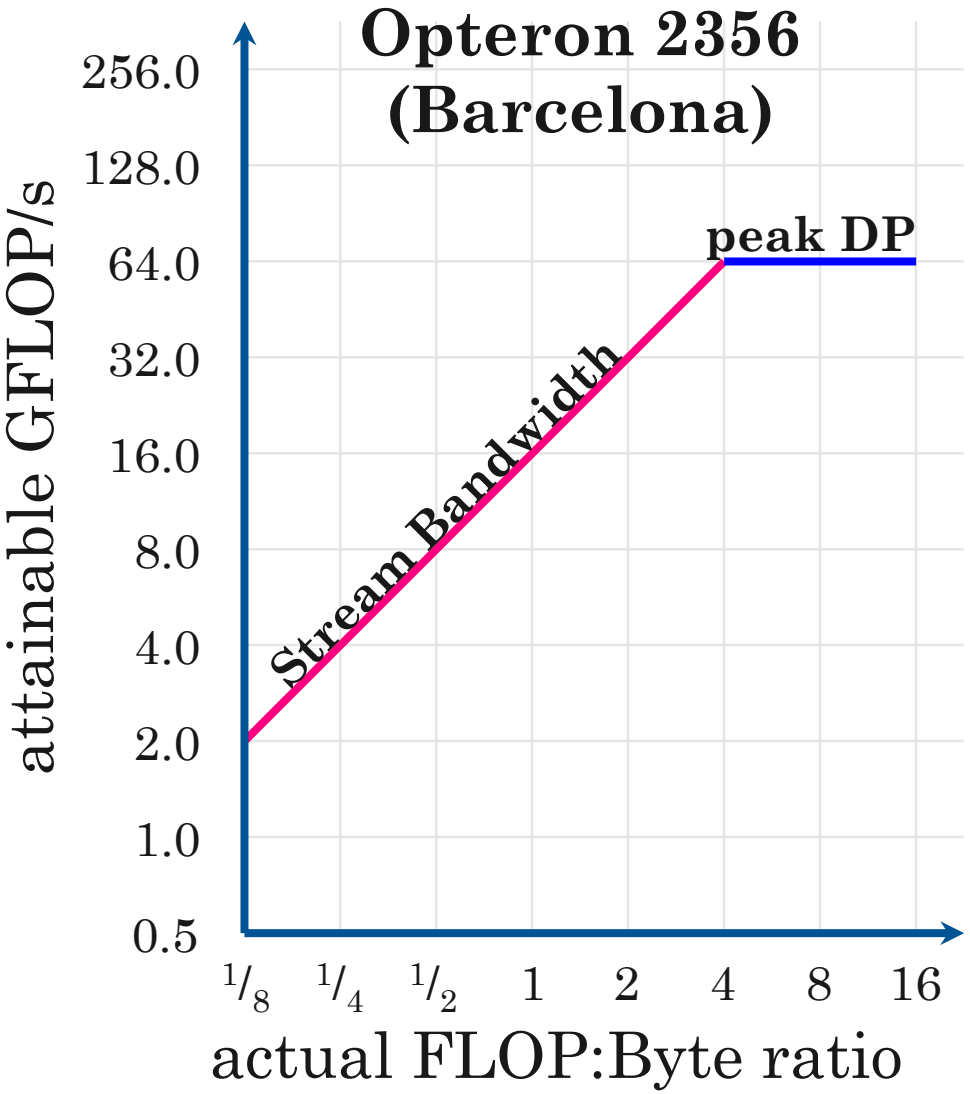



- On Opterons, floating-point instructions have a 4-cycle latency.
- If we don't express 4-way ILP (instruction level parallelism), performance will drop by as much as 4x

# Communication Ceilings

# Roofline Model

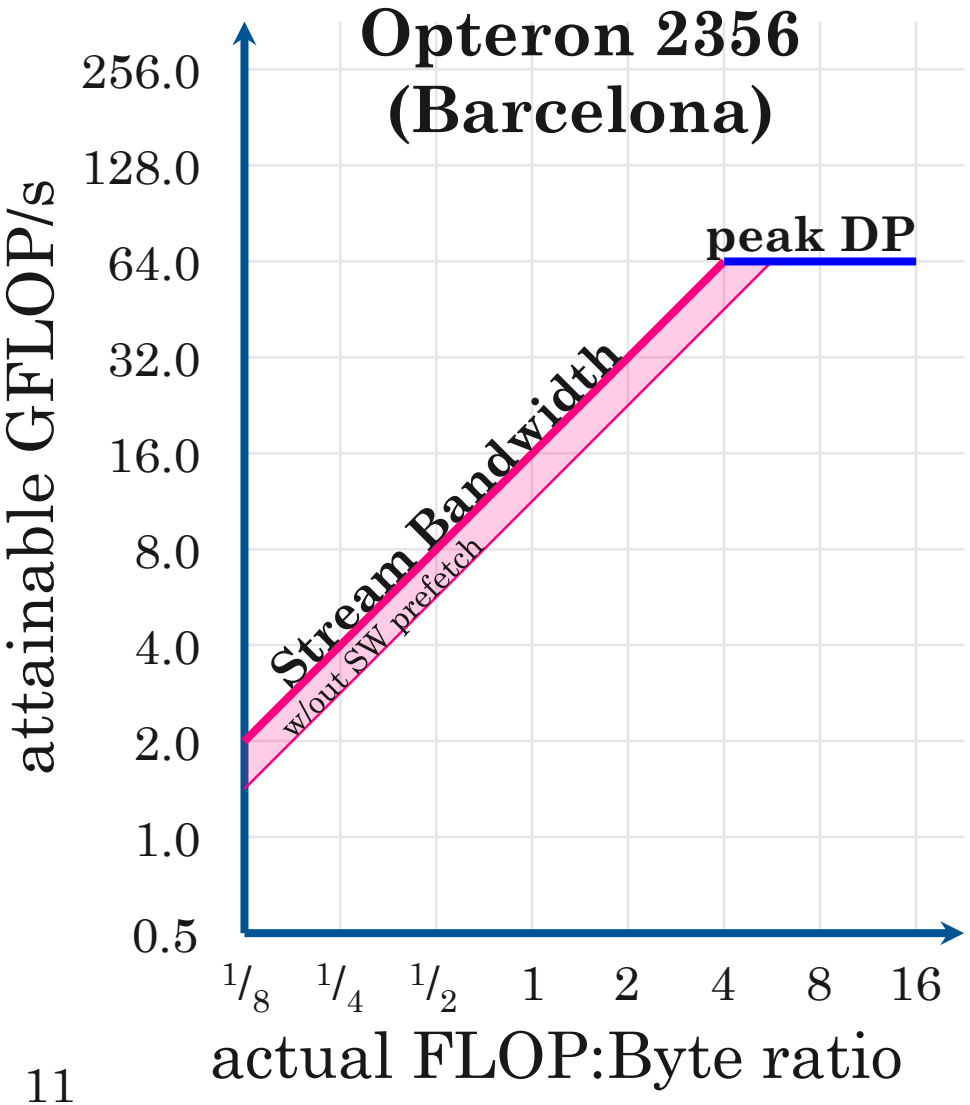
## Communication Ceilings




- We can perform a similar exercise taking away parallelism from the memory subsystem

# Roofline Model

## Communication Ceilings




- Explicit software prefetch instructions are required to achieve peak bandwidth (SW = software)



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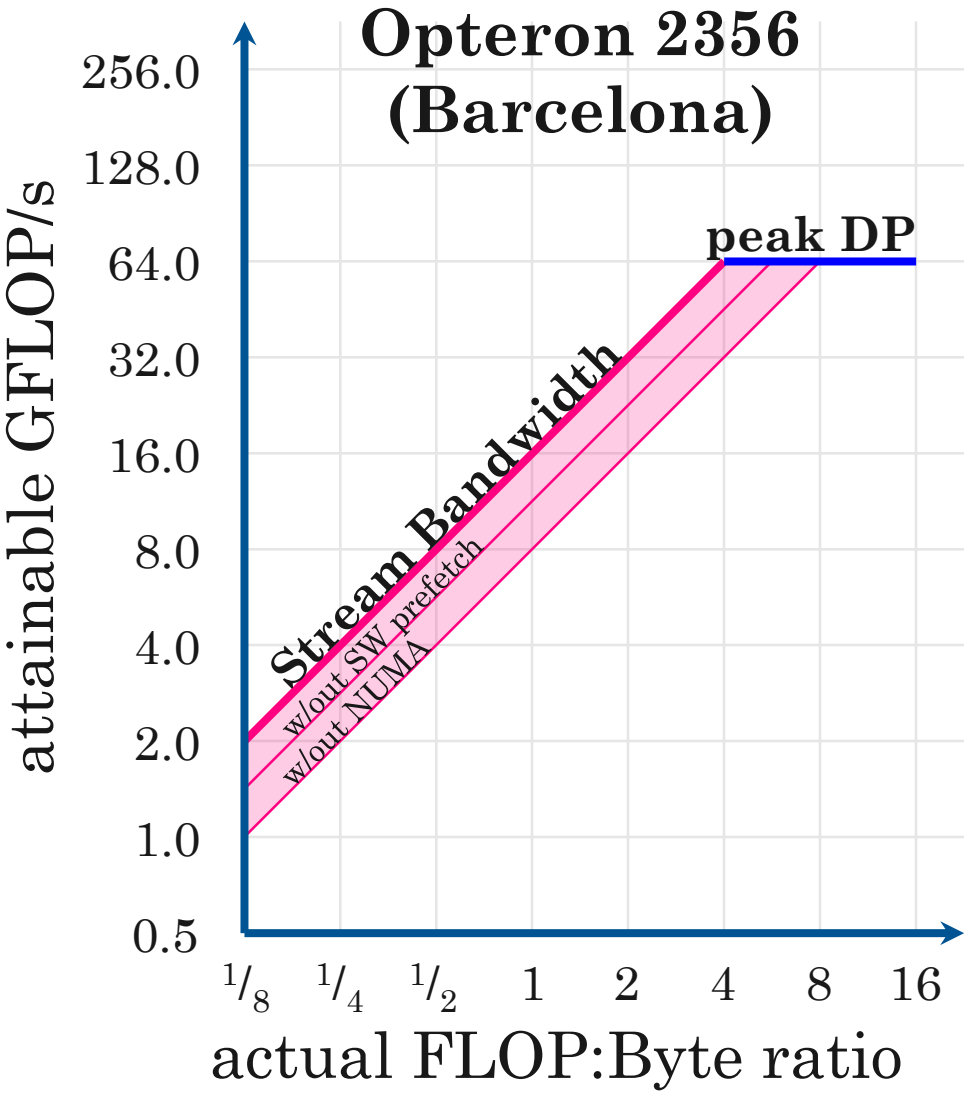
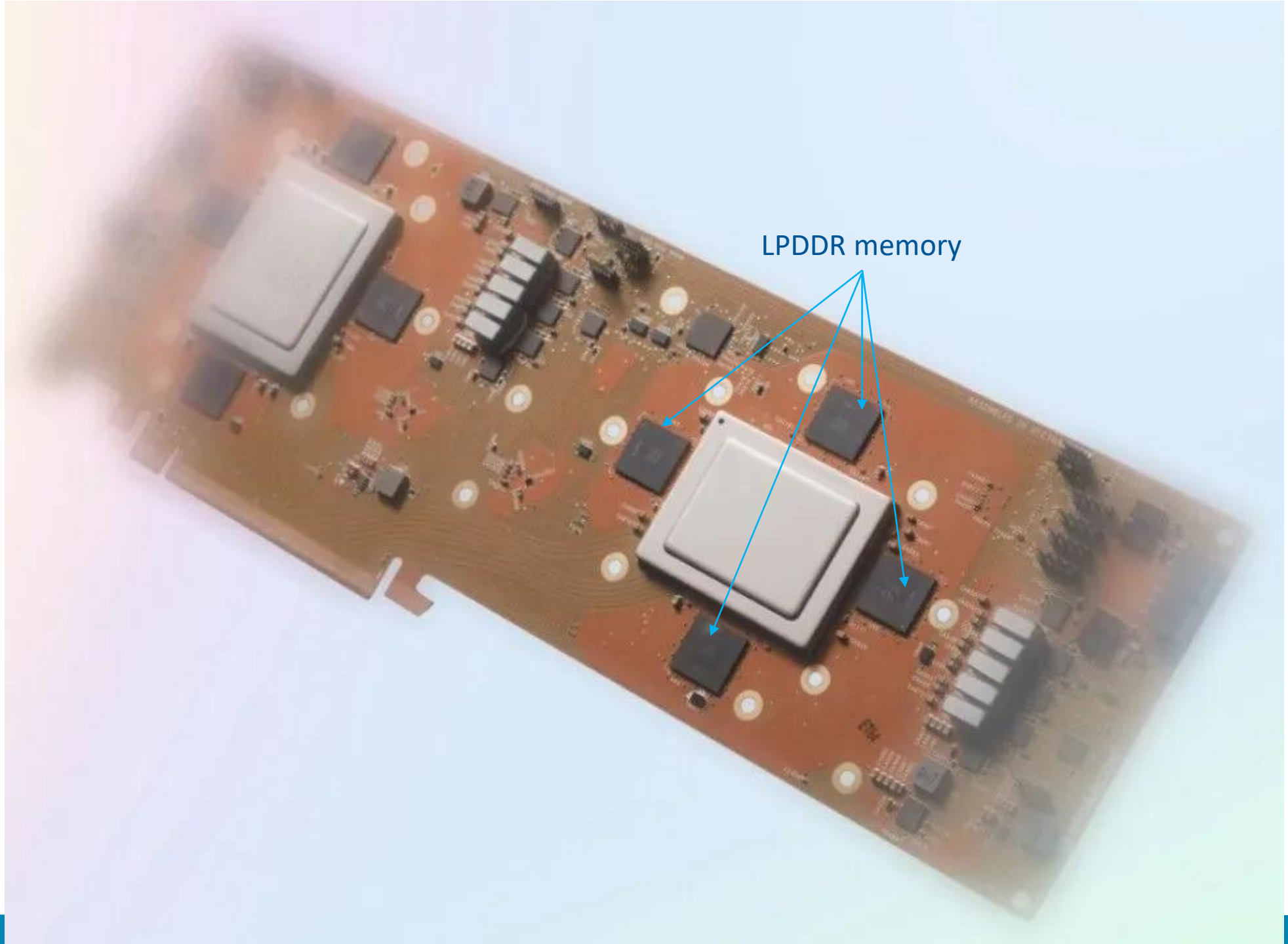
# Roofline Model

## Communication Ceilings



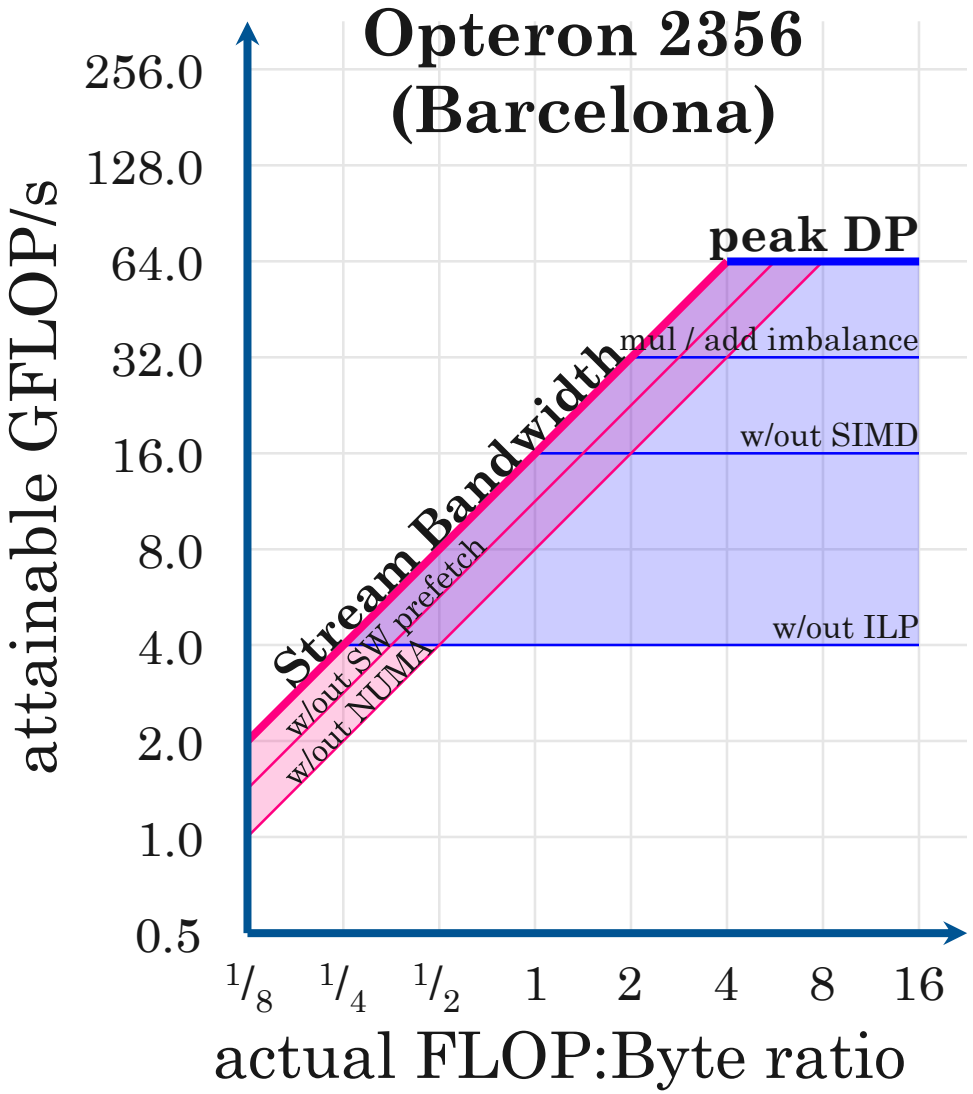

- Opterons are NUMA
- As such memory traffic must be correctly balanced among the two sockets to achieve good Stream bandwidth.
- We could continue this by examining strided or random memory access patterns

MTIA processor
(from Meta)



# Roofline Model

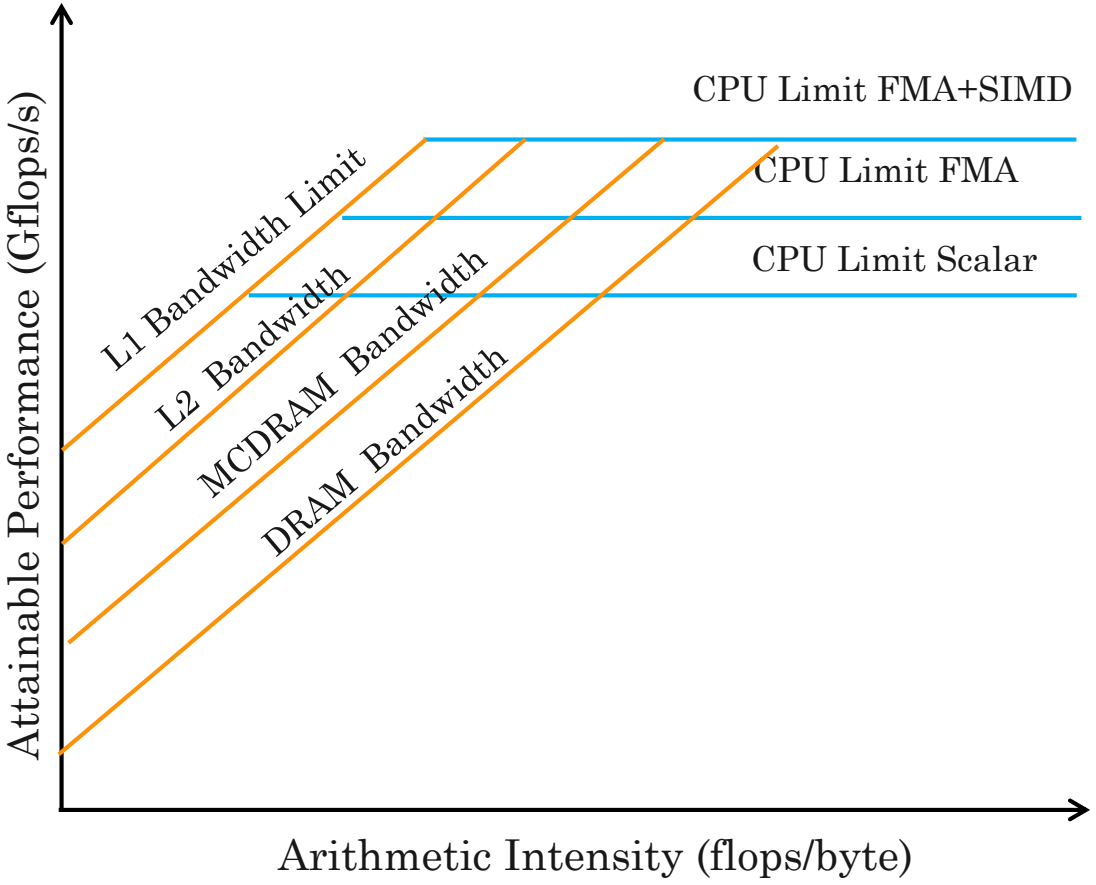
## Computation + Communication Ceilings




- We may bound performance based on the combination of expressed in-core parallelism and attained bandwidth.

# Let’s Take an Example of a Processor: Intel’s Xeon Phi 7250 (Knights Landing or KNL)

# There Are Different Ceilings




KNL theoretical peak performance = Frequency (1.4 GHz) x 8 (vector width) x 2 (dual VPUs) x 2 (FMA inst.) x 68 (number of Cores) = 3.0 TFlops/s

# Numerical Example

- KNL has 3.05 TFLOPS peak double-precision
- 1.4 GHz × 68 core × 8 SIMD × 2 VPUs × 2 FMA = 3046.4 GFLOPS

- AVX frequency is only 1.2 GHz and might throttle down under heavy load (High-AVX code is run only at 1.2GHz)
- ⇒ actual peak: 2611.2 GFLOPS

# Numerical Example

- Without instruction level parallelism (ILP), that is, dual VPUs and FMA
- 1.2 GHz × 68 core × 8 SIMD = 652.8 GFLOPS

- Without ILP, and without SIMD
- 1.2 GHz × 68 core = 84.6 GFLOPS

# Summary: Key Lessons

Computation and communication ceilings

Example of KNL to understand these ceilings numerically